

Simulation-Based Inference for an Adaptive-Network Epidemic Model

April 19, 2026

Millan Bhardwaj

1 Introduction

Infectious disease transmission often depends not only on biological parameters, but also on the structure of the contact network through which infection spreads. In this project, we consider a stochastic SIR epidemic model on an adaptive network, where nodes represent individuals and edges represent potentially infectious contacts. Each individual is in one of three states: susceptible (S), infected (I) or recovered (R). At each discrete time step infection spreads across susceptible–infected edges with transmission probability β , infected nodes recover with probability γ and susceptible nodes may rewire away from infected neighbours with probability ρ . This rewiring mechanism models behavioural adaptation whereby individuals avoid infectious contacts while maintaining connectivity. The model is therefore governed by the parameter vector $\theta = (\beta, \gamma, \rho)$.

The observed data consist of simulator-generated epidemic outputs produced under unknown parameter values. In particular, we observe the infected fraction over time, the number of rewiring events over time and the final degree histogram of the network. The aim of this project is to infer the unknown parameters θ from these observations.

Let y denote the observed data. The goal is to characterise the likelihood $p(y \mid \theta)$ and perform inference on θ . However, this likelihood is not available in closed form.

If the full latent process were observed, including the network configurations $G_{0:T}$ and node states $X_{0:T}$, the model would admit a Markov factorisation:

$$p(G_{0:T}, X_{0:T} \mid \theta) = p(G_0, X_0) \prod_{t=0}^{T-1} p(G_{t+1}, X_{t+1} \mid G_t, X_t, \theta). \quad (1)$$

In practice, only partial summaries $y = s(G_{0:T}, X_{0:T})$ are observed so the likelihood requires marginalising over all latent trajectories:

$$p(y \mid \theta) = \sum_{(G_{0:T}, X_{0:T}): s(\cdot)=y} p(G_{0:T}, X_{0:T} \mid \theta). \quad (2)$$

This summation is computationally intractable due to the large number of possible epidemic and network evolution paths. Consequently, exact likelihood-based inference is not feasible. Instead, I will use simulation-based inference methods.

2 Exploratory Data Analysis

2.1 Mean Infected Fraction and Average Rewiring Activity over Time

We will begin by examining the observed data to understand the dynamics of the epidemic, rewiring behavior and the resulting network structure. This will motivate the choice of summary statistics for the Approximate Bayes Computation procedure.



(a) Mean infected fraction with standard deviation band.

(b) Average rewiring activity with standard deviation band

Figure 1: epidemic and rewiring process over time

In Figure 1 (a) the infected fraction rapidly rises to a peak around $t \approx 9$ followed by a smooth decay to zero. The height and timing of this peak are primarily governed by the transmission and recovery parameters β and γ , higher transmission leads to earlier and larger peaks, while higher recovery suppresses both the peak magnitude and the duration of the epidemic. The variability across replicates is concentrated around the peak, suggesting that statistics such as peak infection and time to peak capture most of the information relevant to β and γ .

(b) shows that rewiring activity is concentrated in an earlier and narrower time window, peaking around $t \approx 5$. Since rewiring is driven by susceptible–infected edges, it is most active during the initial growth phase of the epidemic. The magnitude and timing of this peak provide direct information about the rewiring parameter ρ which cannot be reliably inferred from infection trajectories alone. The relatively small variability outside the peak suggests that most of the informative signal about ρ is concentrated in this short time window.

2.2 Infected Fraction vs Rewiring Count

Figure 2 reveals a nonlinear relationship between infection prevalence and rewiring activity. Rewiring is low when infection is small, increases as infection spreads and then declines again, forming a characteristic hump-shaped pattern consistent with it being driven by susceptible–infected edges, whose number is approximately proportional to $I(t)(1 - I(t))$.

This demonstrates that rewiring is tightly coupled to the infection dynamics. Rewiring peaks earlier than infection during the growth phase when both susceptible and infected individuals are abundant and declines as susceptibles are depleted.

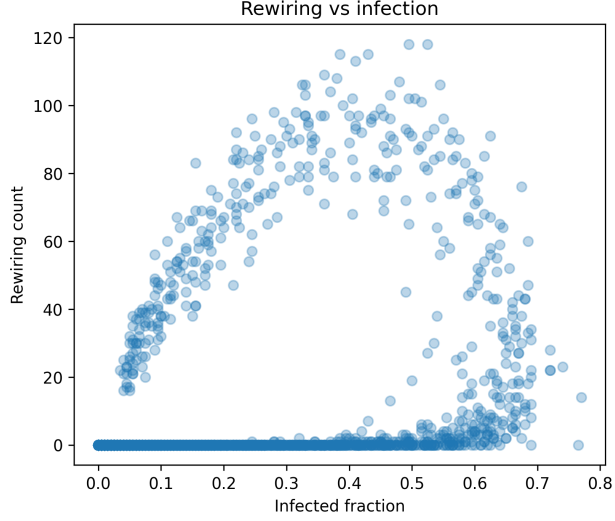


Figure 2: Relationship between infection prevalence and rewiring activity.

The large cluster of points with zero rewiring at nonzero infection reflects time points after susceptible depletion, where S-I edges no longer exist despite ongoing infection. This is further evidence that rewiring carries information distinct from the infection trajectory alone.

Importantly, both higher recovery γ and higher rewiring ρ can suppress infection spread but through different mechanisms: recovery removes infected individuals, while rewiring reduces exposure by breaking susceptible–infected edges. As a result, infection-based summaries alone are insufficient for parameter inference since similar infection trajectories can arise from different combinations of γ and ρ .

Therefore, summary statistics must incorporate both infection and rewiring. In particular, the magnitude and timing of the rewiring peak, especially relative to the infection peak, provide key information about ρ and help disentangle its effect from that of γ .

2.3 Final Degree Histogram

Figure 3 shows that the final degree distribution remains fairly concentrated, with no evidence of extreme hubs or severe fragmentation. This indicates that rewiring primarily reshapes connections locally without significantly altering the overall level of connectivity.

As a result, simple summaries such as the mean and variance are sufficient to capture its key features and are particularly informative about the rewiring parameter ρ , which directly affects network structure.

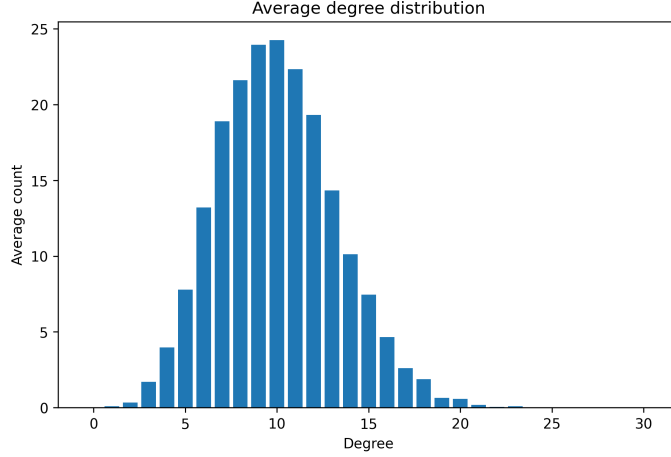


Figure 3: Average final degree distribution of the network.

3 Approximate Bayesian Computation

3.1 ABC Procedure

A total of 100,000 parameter values were sampled from the prior distributions ($\beta \sim \text{Uniform}(0.05, 0.50)$, $\gamma \sim \text{Uniform}(0.02, 0.20)$, $\rho \sim \text{Uniform}(0, 0.8)$). For each draw, the model was simulated and summary statistics were computed. A normalised Euclidean distance between simulated and observed summaries was used, and parameters with the smallest distances were retained to approximate the posterior.

A quantile-based tolerance was used, retaining the top 5%, 2%, and 1% of simulations. Lower tolerances produced more concentrated posteriors. 2% was selected for the main analysis as a balance between accuracy and stability, with 1% and 5% used as sensitivity checks.

A single simulation was used per parameter draw, introducing some noise into the distance metric, though this is mitigated by the large number of simulations.

3.2 Posterior Distributions (Using Set A)

As an initial approach I constructed summary statistics using only the infection time series. Specifically, the peak infected fraction, time to peak and the total infected (area under the curve). I will refer to this initial choice of summary statistics as **Set A**

These statistics are expected to be informative for the infection rate β and the recovery rate γ . However, since the rewiring parameter ρ primarily affects the network structure, its influence is only indirectly reflected in the infection dynamics. As a result, summary statistics based solely on the infection time series are insufficient to fully identify ρ .

The posterior summaries for the selected tolerance level are as follows:

- **Infection rate (β)**
Mean: 0.198, Median: 0.187, 95% CI: (0.096, 0.353)

- **Recovery rate (γ)**
Mean: 0.0824, Median: 0.0820, 95% CI: (0.0658, 0.1000)
- **Rewiring rate (ρ)**
Mean: 0.412, Median: 0.417, 95% CI: (0.0245, 0.783)

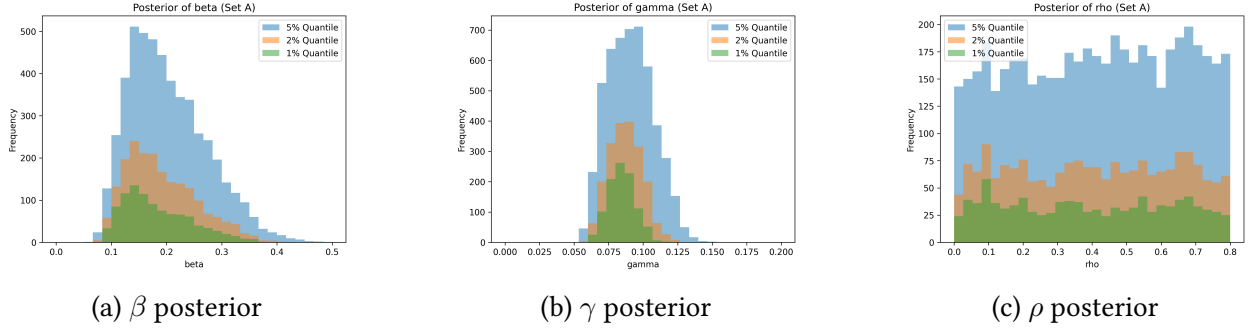


Figure 4: Approximate posterior distributions for parameters β , γ , and ρ under Set A for different tolerance levels.

These results clearly demonstrate the differences in parameter identifiability. The recovery rate γ is well identified, with a tightly concentrated posterior distribution that remains stable across tolerance levels. The infection rate β is also reasonably well identified, although with greater uncertainty reflected in a wider credible interval.

In contrast, the posterior distribution of ρ remains diffuse and spans a large portion of the parameter space. This indicates that the chosen statistics focusing on infection-based summaries do not sufficiently capture the influence of rewiring on the system. This is consistent with previous observations that rewiring primarily affects network structure rather than infection dynamics, motivating the inclusion of additional network-based summary statistics in the next stage.

3.3 Posterior Distributions (Set B)

To address the limitations of Set A, **Set B** introduces additional summary statistics from the rewiring time-series: peak rewiring activity, time to peak rewiring and total rewiring. All summary statistics were standardised prior to computing distances to ensure balanced contributions. $N = 150,000$ simulations were used to compensate for the increased dimensionality of the summary statistics.

The posterior summaries for the selected tolerance level (2%) are as follows:

- **Infection rate (β)**
Mean: 0.177, Median: 0.174, 95% CI: (0.107, 0.264)
- **Recovery rate (γ)**
Mean: 0.0980, Median: 0.0965, 95% CI: (0.0584, 0.147)

- **Rewiring rate (ρ)**

Mean: 0.306, Median: 0.304, 95% CI: (0.178, 0.446)

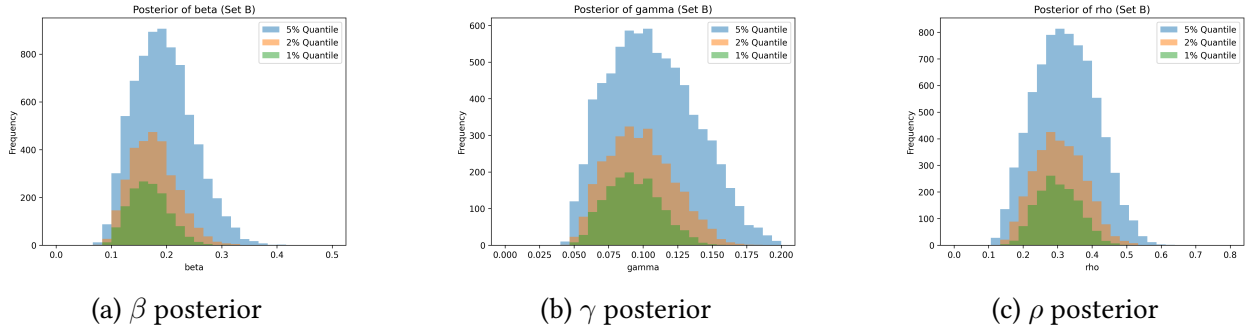


Figure 5: Approximate posterior distributions for parameters β , γ , and ρ under Set B for different tolerance levels.

the posteriors for β and γ are well concentrated, indicating that these parameters are reliably identified. γ exhibits a slight increase in uncertainty from set A (due to a trade-off with ρ) and β shows a clear central region with uncertainty decreasing from set A

The most notable improvement is observed for ρ . Compared to Set A its posterior is substantially more concentrated, with a clear peak around intermediate values. This demonstrates that the inclusion of rewiring-based summary statistics successfully captures information about the adaptive network dynamics and improves identifiability of ρ .

However, the posterior for ρ remains significantly wider than those of β and γ , indicating that uncertainty in this parameter is still relatively high. This suggests that, while rewiring summaries provide useful information, they do not fully capture the structural effects of rewiring on the network.

3.4 Posterior Distribution (Set C)

To address the limitation of Set B relying on time-series summaries and failing to fully capture the underlying network structure, **Set C** incorporates statistics from the final degree distribution, adding: the Mean degree, variance of degrees and high-degree mass (proportion of nodes with degree ≥ 15 to capture the distribution shape of the histogram).

$N = 200,000$ simulations were used to account for the higher-dimensional summary vector. Distances between simulated and observed summaries were computed using a normalised Euclidean distance, where each summary statistic was scaled by its empirical standard deviation.

Here are the results, focusing on the 2% quantile for consistency:

- **Infection rate (β)**

Mean: 0.191, Median: 0.186, 95% CI: (0.106, 0.303)

- **Recovery rate (γ)**

Mean: 0.106, Median: 0.103, 95% CI: (0.056, 0.174)

- **Rewiring rate (ρ)**

Mean: 0.315, Median: 0.314, 95% CI: (0.164, 0.471)

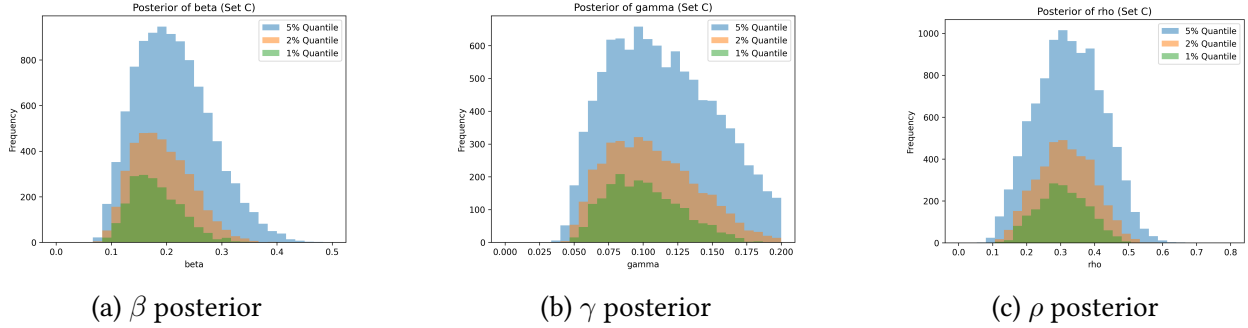


Figure 6: Approximate posterior distributions for parameters β , γ , and ρ under Set C for different tolerance levels.

The posterior summaries in Set C are very similar to those obtained in Set B, with comparable means, medians and credible intervals. This consistency suggests that the inferred parameters are robust to the choice of summary statistics and thus accurate.

However, the posterior distributions in Set C are slightly wider for all parameters. This reflects the curse of dimensionality in the distance metric.

In particular, the uncertainty in γ increases from Set A to Set C. This is likely due to a trade-off with ρ , as both parameters influence the decline of the epidemic. The inclusion of network-based summaries provides more direct information about ρ , allowing greater flexibility in how γ and ρ jointly explain the dynamics and hence increasing uncertainty in γ .

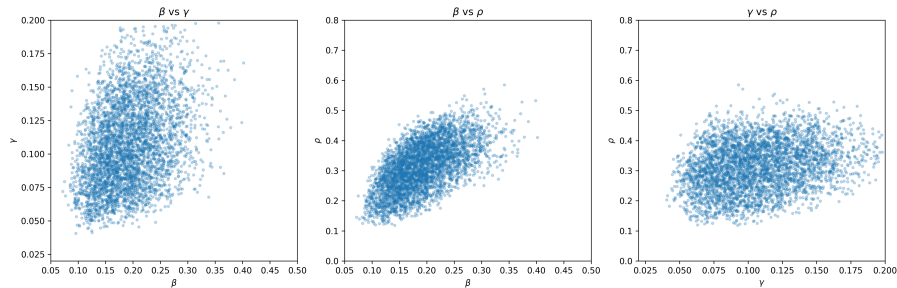


Figure 7: Joint posterior distributions for β , γ , and ρ obtained using rejection ABC (Set C) at the 2% tolerance level.

As shown in Figure 7, the joint posterior distributions reveal the dependence structure between parameters. A strong positive association is observed between β and ρ and a weaker positive association between β and γ , reflecting the need to offset higher transmission rates with increased recovery or rewiring. In contrast, the relationship between γ and ρ appears weak. This suggests that the inclusion of network-based summaries in Set C helps distinguish the effects of recovery and behavioural adaptation, reducing potential confounding.

3.5 Robustness Check for Set C

Table 1: Comparison of posterior summaries for Set C under different runs and distance metrics: mean (median) [95% CI]

	Original Run	New Seed	L1 Distance (New Seed)
β	0.191 (0.186) [0.106, 0.303]	0.193 (0.189) [0.105, 0.306]	0.189 (0.184) [0.104, 0.304]
γ	0.106 (0.103) [0.056, 0.174]	0.107 (0.104) [0.057, 0.173]	0.106 (0.101) [0.055, 0.179]
ρ	0.315 (0.314) [0.164, 0.471]	0.315 (0.316) [0.164, 0.463]	0.315 (0.315) [0.154, 0.474]

Table 1 shows that posterior summaries are highly consistent across runs and distance metrics, with negligible differences in means, medians and credible intervals. This indicates that the inference is stable to both Monte Carlo variability and the choice of distance metric.

3.6 ABC rejection conclusion

Across the three sets, progressively richer summary statistics improved inference, particularly for the rewiring parameter ρ , although uncertainty remained relatively high.

However, rejection ABC has several limitations. First, it approximates $p(\theta \mid S_{\text{obs}})$ rather than the true posterior $p(\theta \mid y_{\text{obs}})$, leading to loss of information due to the use of summary statistics. Second, the tolerance is defined via a distance metric that is not probabilistically grounded, which can introduce bias in the posterior approximation. Third, the method is computationally inefficient as acceptance rates decrease rapidly with tighter tolerances and higher-dimensional summaries, requiring a large number of simulations.

These limitations motivate the use of more advanced simulation-based methods including Regression-adjusted ABC and Sequential Monte Carlo ABC (SMC-ABC).

4 Regression-Adjusted ABC

4.1 Motivation

Although Set C improves inference by incorporating network structure, rejection ABC remains approximate because accepted simulations satisfy $s_i \approx s_{\text{obs}}$ rather than exact equality. This leaves residual discrepancies between simulated and observed summaries. The problem becomes more pronounced in higher-dimensional settings such as Set C. To reduce this error, we apply the regression adjustment method [BZB02].

4.2 Method

The regression adjustment assumes that, locally, the relationship between parameters and summary statistics is approximately linear:

$$\theta_i = \alpha + \beta(s_i - s_{\text{obs}}) + \epsilon_i.$$

Using the accepted samples, this relationship is estimated by linear regression. Each parameter value is then adjusted according to:

$$\theta_i^{\text{adj}} = \theta_i - \hat{\beta}(s_i - s_{\text{obs}}),$$

which projects the parameter onto the value it would take if the simulated summaries exactly matched the observed summaries.

This procedure is applied as a post-processing step to the accepted samples and therefore requires no additional simulations. To ensure validity, adjusted parameter values are truncated to lie within the support of their prior distributions.

4.3 results

The regression-adjusted posterior distributions at the 2% tolerance level are summarised below:

- **Infection rate (β)**
Mean: 0.141, Median: 0.139, 95% CI: (0.094, 0.202)
- **Recovery rate (γ)**
Mean: 0.085, Median: 0.084, 95% CI: (0.066, 0.113)
- **Rewiring rate (ρ)**
Mean: 0.287, Median: 0.286, 95% CI: (0.232, 0.349)

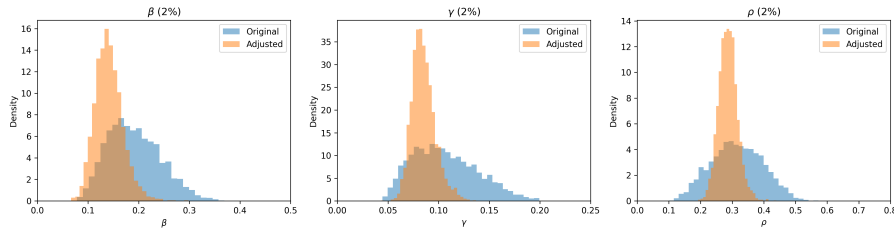


Figure 8: Comparison of rejection ABC and regression-adjusted posterior distributions for β , γ , and ρ at the 2% tolerance level.

Relative to the original rejection ABC results, the regression-adjusted posteriors are substantially more concentrated. The most notable improvement is observed for the rewiring parameter ρ , whose 95% credible interval shrinks from (0.164, 0.471) to (0.232, 0.349).

In addition, the posterior means for all parameters shift downward following adjustment:

$$\beta : 0.191 \rightarrow 0.141, \quad \gamma : 0.106 \rightarrow 0.085, \quad \rho : 0.315 \rightarrow 0.287.$$

The downward shift in posterior means following adjustment is consistent with tolerance induced upward bias in rejection ABC, whereby accepted samples systematically overestimate parameters due to the finite acceptance region.

The regression-adjusted joint posteriors (Figure 9) retain the dependence structure of rejection ABC but with reduced dispersion, indicating that the adjustment contracts the posterior without distorting parameter relationships.

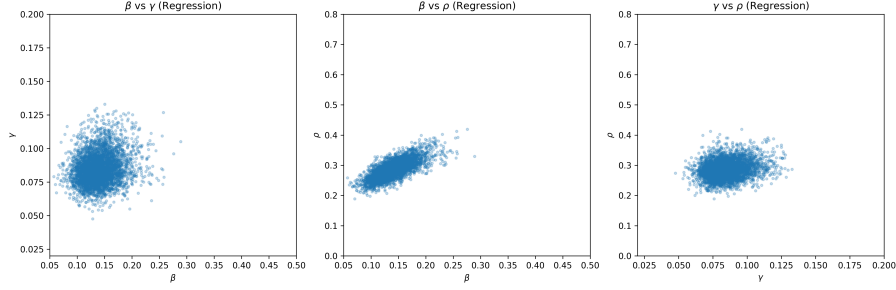


Figure 9: Joint posterior distributions for β , γ , and ρ obtained using regression-adjusted ABC (Set C) at the 2% tolerance level.

4.4 Synthetic Truth Recovery Experiment

To assess calibration a synthetic dataset is simulated at $\theta^* = (0.20, 0.10, 0.30)$ and the Set C pipeline is applied treating this as observed data.

For rejection ABC (2%), posterior means are $\beta = 0.266$ (95% CI: 0.144–0.431), $\gamma = 0.123$ (0.070–0.188), and $\rho = 0.306$ (0.171–0.457), indicating upward bias and substantial dispersion.

Regression adjustment yields $\beta = 0.212$ (0.142–0.299), $\gamma = 0.111$ (0.093–0.136), and $\rho = 0.295$ (0.243–0.352), with reduced variance and posterior means closer to the true values. In all cases, the true parameters lie within the corresponding 95% credible intervals.

These results indicate that rejection ABC suffers from tolerance-induced bias, while regression adjustment effectively corrects this and improves precision without compromising coverage, demonstrating good calibration of the inference procedure.

4.5 Regression-Adjusted ABC Conclusion

The results show that regression adjustment improves posterior inference by reducing dispersion while preserving the underlying dependence structure between parameters. This is particularly evident for ρ , where the adjusted posterior is more concentrated while remaining consistent with the rejection ABC results. This improvement is also reflected in the synthetic truth recovery experiment, where regression adjustment yields posterior estimates closer to the true parameter values while maintaining correct coverage.

However, the method relies on a local linear approximation between parameters and summary statistics. As the relationship between ρ and the epidemic dynamics is highly nonlinear (Figure 2), this limits the accuracy of the correction, particularly for samples further from s_{obs} . Although the synthetic truth recovery experiment suggests that in practice this impact is limited.

Moreover, regression adjustment does not address the fundamental limitation of ABC as inference is still based on summary statistics rather than the full data, leading to unavoidable information loss. It also does not improve sampling efficiency, since it depends on the initial rejection step and therefore still requires a large number of simulations.

5 Sequential Monte Carlo ABC (SMC-ABC)

5.1 Motivation

While regression-adjusted ABC reduces bias, it does not address the inefficiency of rejection ABC, where many simulations are wasted sampling from low-probability regions of the prior.

SMC-ABC overcomes this by sampling from a sequence of intermediate distributions that progressively concentrate around the posterior, focusing computation on high-probability regions and improving efficiency.

5.2 Method

[SFT07] The SMC-ABC algorithm proceeds through a sequence of populations with decreasing tolerances $\epsilon_1 > \epsilon_2 > \dots > \epsilon_T$.

At each iteration particles are resampled according to their weights, perturbed using a Gaussian kernel, and accepted if the simulated summaries satisfy the tolerance criterion.

The proposal kernel covariance is adapted as

$$\Sigma_t = 2 \text{Cov}_w(\theta^{(t-1)}) \quad [\text{BCMR09}]$$

allowing the algorithm to transition from broad exploration to local refinement.

Importance weights correct for sampling from a proposal distribution rather than the prior, ensuring that the final particle population approximates the ABC posterior.

The SMC-ABC algorithm was implemented with $N = 2,000$ particles and a tolerance schedule

$$\epsilon = (1.6825, 1.4680, 1.2383, 1.0901)$$

corresponding approximately to the 10%, 5%, 2%, and 1% quantiles of the distance distribution.

The total number of simulations required was 85,853. The effective sample size (ESS) remained high throughout, reaching approximately 1800 in the final population, indicating minimal particle degeneracy and stable weighting.

For comparison, rejection ABC in Set C required 200,000 simulations to obtain 2,000 samples at the 1% tolerance level. SMC-ABC therefore achieved a posterior of comparable effective size using less than half the number of simulations, demonstrating substantially improved computational efficiency.

5.3 Results

The final SMC-ABC posterior estimates reported here are based on a single run of the sequential Monte Carlo algorithm with the chosen tolerance schedule and particle population:

- **Infection rate (β)**

Mean: 0.179, Median: 0.176, 95% CI: (0.102, 0.277)

- **Recovery rate (γ)**
Mean: 0.099, Median: 0.097, 95% CI: (0.058, 0.154)
- **Rewiring rate (ρ)**
Mean: 0.307, Median: 0.307, 95% CI: (0.175, 0.452)

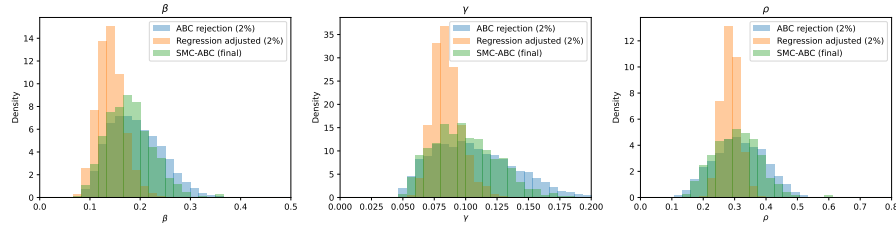


Figure 10: Comparison of posterior distributions obtained using rejection ABC, regression-adjusted ABC, and SMC-ABC at the 2% tolerance level.

The SMC-ABC posteriors are very similar in both location and spread to those obtained using rejection ABC (Set C). This indicates that the inferred posterior is stable across methods and that the chosen summary statistics are highly informative.

Although earlier analysis suggested that the relationship between ρ and the summary statistics may exhibit nonlinear effects, the similarity of posterior distributions across methods indicates that this does not significantly hinder inference with the chosen summary statistics.

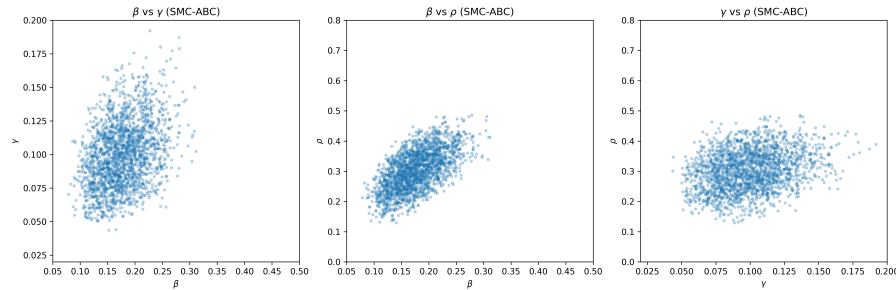


Figure 11: Joint posterior distributions for β , γ , and ρ obtained using SMC-ABC (Set C).

As shown in Figure 11 the SMC-ABC joint posteriors are consistent with those from rejection ABC and regression adjustment, confirming the dependence structure is stable across methods

5.4 Posterior predictive Checks

Posterior predictive simulations from the final SMC-ABC posterior closely reproduce the observed data across all summaries. The infection trajectory, rewiring dynamics and final degree distribution all lie well within the predictive envelope, indicating that the model captures both the epidemic dynamics and the resulting network structure. This provides strong evidence that the inferred posterior gives an adequate description of the data-generating process.

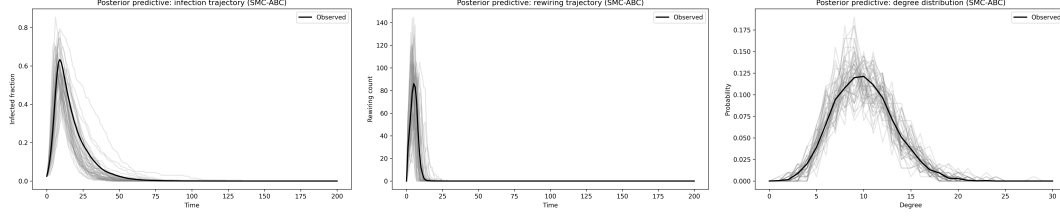


Figure 12: Posterior predictive checks under the final SMC-ABC posterior. For 50 draws of $\theta = (\beta, \gamma, \rho)$, the model was simulated once per draw.

5.5 SMC-ABC conclusion

SMC-ABC produces posterior distributions that closely match those obtained using both rejection ABC and regression-adjusted ABC. However, SMC-ABC achieves this with substantially improved efficiency by concentrating sampling in high-probability regions, while maintaining high effective sample sizes and avoiding particle degeneracy. Posterior predictive checks further confirm that the inferred posterior provides a good fit to the observed data.

Compared to regression-adjusted ABC, the SMC-ABC posteriors are slightly wider, reflecting the absence of a linear correction and making it a more conservative and principled approach.

Despite these advantages, SMC-ABC remains subject to the limitations of summary-based inference which include: information loss and dependence on the tolerance level. Furthermore, it is more complex to implement, requiring careful tuning of tolerances and proposal kernels.

6 Conclusion

In this project, simulation-based inference methods were used to estimate the parameters of an adaptive-network epidemic model. Across approaches, the inferred values are approximately $\beta \approx 0.14\text{--}0.19$, $\gamma \approx 0.085\text{--}0.106$, and $\rho \approx 0.287\text{--}0.315$.

Increasing the richness of summary statistics improved parameter identifiability, particularly for ρ , with network-based summaries playing a key role in separating its effect from γ . Posterior estimates were broadly consistent across methods, indicating that the results are robust to the choice of inference procedure. Regression adjustment reduced bias, as demonstrated by synthetic truth recovery, suggesting that the true parameter values lie toward the lower end of these ranges, while SMC-ABC achieved similar posterior locations with substantially improved efficiency.

Despite these improvements, inference remains approximate and dependent on the choice of summary statistics. A natural extension would be to combine SMC-ABC with regression adjustment on the final particle population to jointly improve efficiency and reduce bias.

References

- [BCMR09] Mark A. Beaumont, Jean-Marie Cornuet, Jean-Michel Marin, and Christian P. Robert. Adaptive approximate bayesian computation. *Biometrika*, 96(4):983–990, 2009.

- [BZB02] Mark A. Beaumont, Wenyang Zhang, and David J. Balding. Approximate bayesian computation in population genetics. *Genetics*, 162(4):2025–2035, 2002.
- [SFT07] Scott A. Sisson, Yanan Fan, and Mark M. Tanaka. Sequential monte carlo without likelihoods. *Proceedings of the National Academy of Sciences*, 104(6):1760–1765, 2007.

